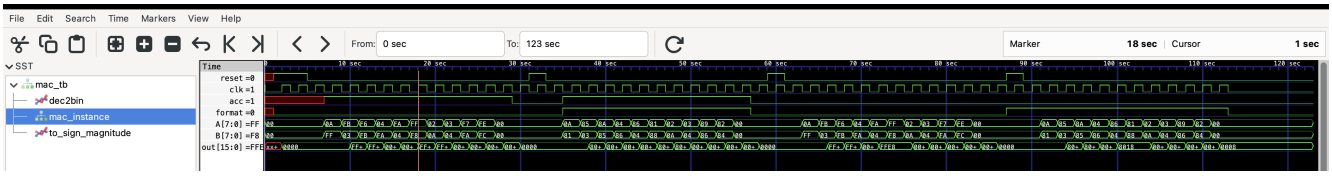
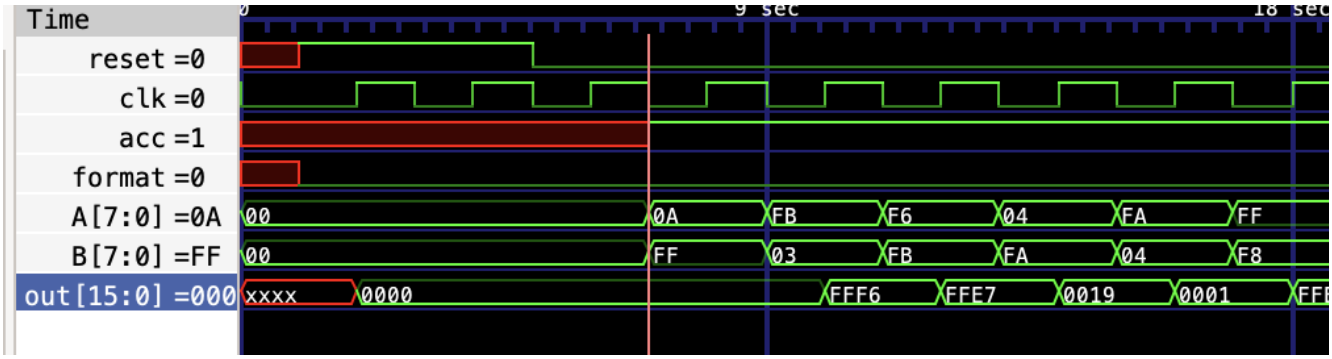
ECE 284 HW 1
Reconfigurable MAC Unit

Russ Sobti

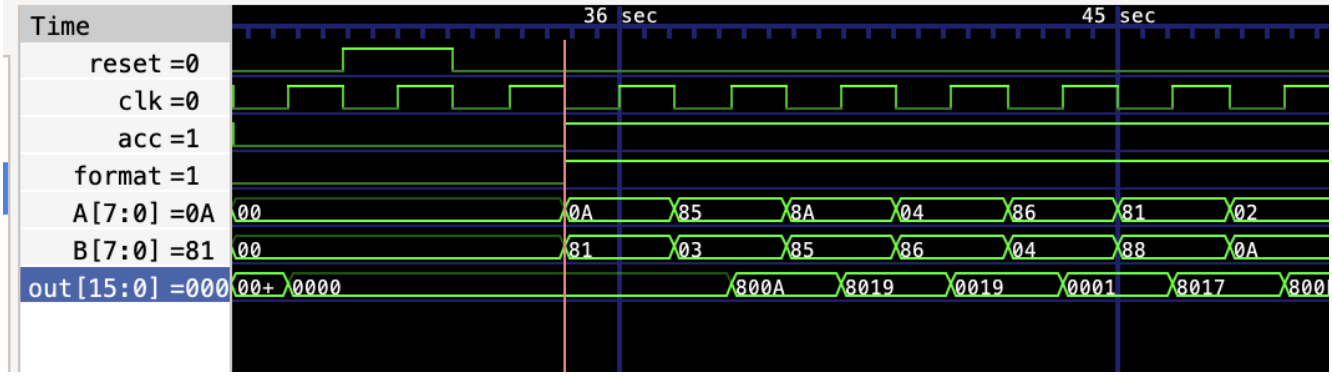
1 Overall



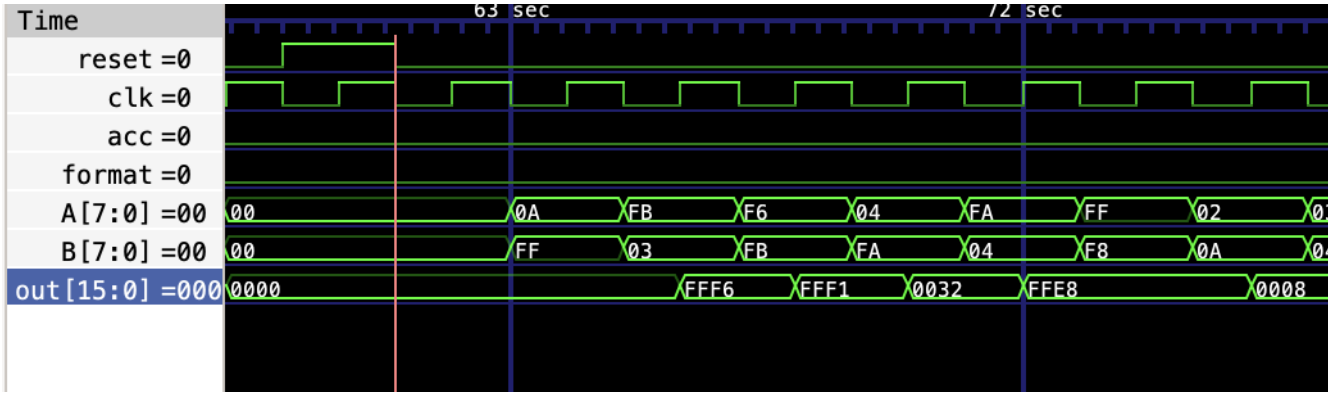
2 Format 0, Acc 1



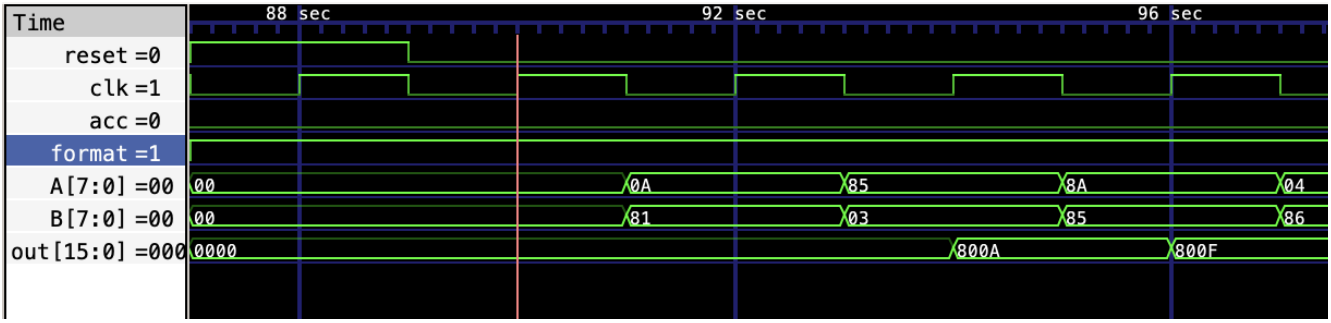
3 Format 1, Acc 1



4 Format 0, Acc 0



5 Format 1, Acc 0



6 Code Snippet

```
reg signed [psum_bw-1:0] psum_q;
reg signed [bw-1:0] a_q;
reg signed [bw-1:0] b_q;

wire [psum_bw-1:0] psum_magnitude = psum_q < 0 ? -psum_q : psum_q;
assign out = format ? {psum_q[psum_bw-1], psum_magnitude[psum_bw-2:0]} : psum_q;

// Your code goes here
always @(posedge clk) begin
        if(format) begin
                a_q <= A[bw-1] ? -$signed({1'b0, A[bw-2:0]}) : $signed({1'b0, A[bw
    -2:0]});
                b_q <= B[bw-1] ? -$signed({1'b0, B[bw-2:0]}) : $signed({1'b0, B[bw
    -2:0]});
        end else begin
                a_q <= A;
                b_q <= B;
        end
        if (reset) begin
                psum_q <= 0;
        end
        else if(acc == 1) begin
                psum_q <= psum_q + (a_q * b_q);
        end
        else begin
                psum_q <= (a_q * b_q);
        end
end
```

7 Linux Cloud Realpath

/home/linux/ieng6/ECE284_FA26_001/rsobti/ece284fa26/hardware/wk1/hw1